

Case Study

Duration- 90 Minutes

A global enterprise wants to build an **Agentic AI-powered IT Operations Platform** capable of autonomously managing production incidents.

The system should:

- Monitor alerts from multiple systems
- Analyse logs and telemetry
- Retrieve troubleshooting SOPs
- Identify probable root causes
- Execute approved remediation workflows
- Maintain memory of past incidents
- Coordinate across multiple specialized AI agents
- Interact with enterprise tools using MCP

Example Scenario

A payment application shows:

- High latency
- Increased error rate
- Database timeout exceptions

The AI system should:

1. Detect anomaly
2. Gather logs and metrics
3. Retrieve similar historical incidents
4. Generate root-cause hypothesis
5. Recommend remediation
6. Execute safe automation steps
7. Escalate if confidence is low

Area needs to focus on

Topic

LLM Selection

Prompt Engineering

Vector DB

RAG

Guardrails

Agentic AI

Design Patterns

Short-Term Memory

Topic

Long-Term Memory

MCP

Expected Agents

- Monitoring Agent
- Log Analysis Agent
- RAG Retrieval Agent
- Decision Agent
- Automation Agent
- Compliance Agent
- Escalation Agent

Kindly mention Tech Stack based on your knowledge

Deliveries Expected:

1. End to End Architecture Diagram: 50%
2. How are you handling below scenarios: 30%
 - Why multi-agent?
 - Which design pattern?
 - Memory strategy?
 - Human-in-the-loop boundaries?
 - Tool calling strategy?
 - Failure handling?
 - Cost optimization?
 - Observability of agents?
3. Description of Each layer Mentioned in Diagram and concept used: 20%